MANIDEEP CHITTIMODU

Gen AI / ML Engineer | NLP, LLMs & Applied Machine Learning

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SUMMARY

AI / ML engineer with hands-on experience building and shipping deep-learning, NLP, and generative-AI systems end to end. Built a CNN-based computer-vision security system, an explainable recommendation model using custom similarity scoring, and an LLM assistant grounded in product data via embedding-based semantic search. Comfortable across the ML lifecycle — data preprocessing, feature engineering, training and model evaluation, and deploying the model behind a real interface. B.Tech in Artificial Intelligence & Data Science with a solid mathematical foundation (linear algebra, probability, statistics); IBM Data Science and Qualcomm AI Upskilling certified.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Dart • C, Java (coursework)
Machine Learning: TensorFlow, PyTorch, Scikit-learn, supervised & unsupervised learning, CNNs, deep learning, fine-tuning, model evaluation, feature engineering
Generative AI & NLP: LLMs & GPT (OpenAI API), LangChain, RAG pipelines, vector embeddings, prompt engineering, Hugging Face, NLP, AI agents
Mathematics: Linear algebra, probability, statistics
Data & MLOps: Pandas, NumPy, SQL, ETL / data pipelines, Plotly Dash, Jupyter, model deployment, Git, Docker, CI/CD
Dev & Cloud: FastAPI, Node.js, React.js, Next.js, PostgreSQL, MongoDB, AWS, GCP

EXPERIENCE

Web Development Intern — Prodigy InfoTech, Mumbai *09/2025 – 12/2025*

- Built responsive web pages and front-end features using modern web technologies; strengthened debugging, problem-solving, and clean-code practices on practical development tasks.
- Worked in a professional Agile environment following best practices and deadlines; completed the 3-month internship and received a Letter of Recommendation for outstanding performance.

KEY PROJECTS

Vastra | ***Node.js · Express · React · MongoDB · Hugging Face API · Embeddings*** *2025*

- Built an AI assistant that answers buyer questions from live product data using embedding-based semantic search (retrieval + LLM generation) over a Hugging Face model, plus AI-assisted bulk-quote suggestions for suppliers.
- Delivered the surrounding B2B marketplace with fully isolated buyer/supplier sessions and business rules enforced consistently across UI, cart, and API.

NextCinema | ***Next.js · TypeScript · MongoDB · TMDB API · Framer Motion*** *2025*

- Designed a custom, explainable similarity-scoring recommendation model (genre, cast, director, rating, popularity), prioritising interpretability over a black-box approach.
- Implemented self-hosted JWT session auth, guest browsing, and account-gated personalisation over MongoDB.

Falcon 9 Landing Prediction | ***Python · Pandas · SQL · Scikit-learn · Plotly Dash*** *2025*

- Applied Data Science Capstone (IBM): ingested the SpaceX REST API and scraped records; ran SQL / visual EDA and feature engineering.
- Trained and evaluated Scikit-learn classification models (supervised learning) to predict first-stage landing success; presented results in a Plotly Dash dashboard.

AI-Based Uniform Detection & Authentication | ***Python · OpenCV · DeepFace · CNN · SQLite*** *2025*

- Built a CNN classifier and face-recognition pipeline (OpenCV, DeepFace) for face, ID, and uniform recognition at entry points.
- Owned dataset collection, preprocessing, training, and evaluation; deployed with real-time attendance logging and dress-code compliance reporting.

EDUCATION

VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE — Hyderabad

M.Tech (Pursuing) — Computer Science, CGPA: 8.8/10 *09/2025 – Present*

B.Tech — Artificial Intelligence & Data Science, CGPA: 7.2/10 *09/2021 – 06/2025*

CERTIFICATIONS

- RAG and Agentic AI — IBM / Coursera
- IBM Data Science Professional Certificate — IBM / Coursera
- AI Upskilling Certificate: Hands-On Development from Model to App — Qualcomm Technologies
- End-to-End Machine Learning — Udemy
- Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft
- Cloud Computing (Elite) — NPTEL / IIT

ACHIEVEMENTS & LEADERSHIP

- 1st Prize, IP2K24 Project Expo — VGNT • Organised a 24-hr Hackathon, “Byte Crunchers” (1000+ applicants).
- Department Representative, AIDAYS 2024 • Social Media Head, AI & DS Dept. • NSS Dept. Co-Ordinator — VGNT.